

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I M.Santhosh kumar _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Santhosh _____

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

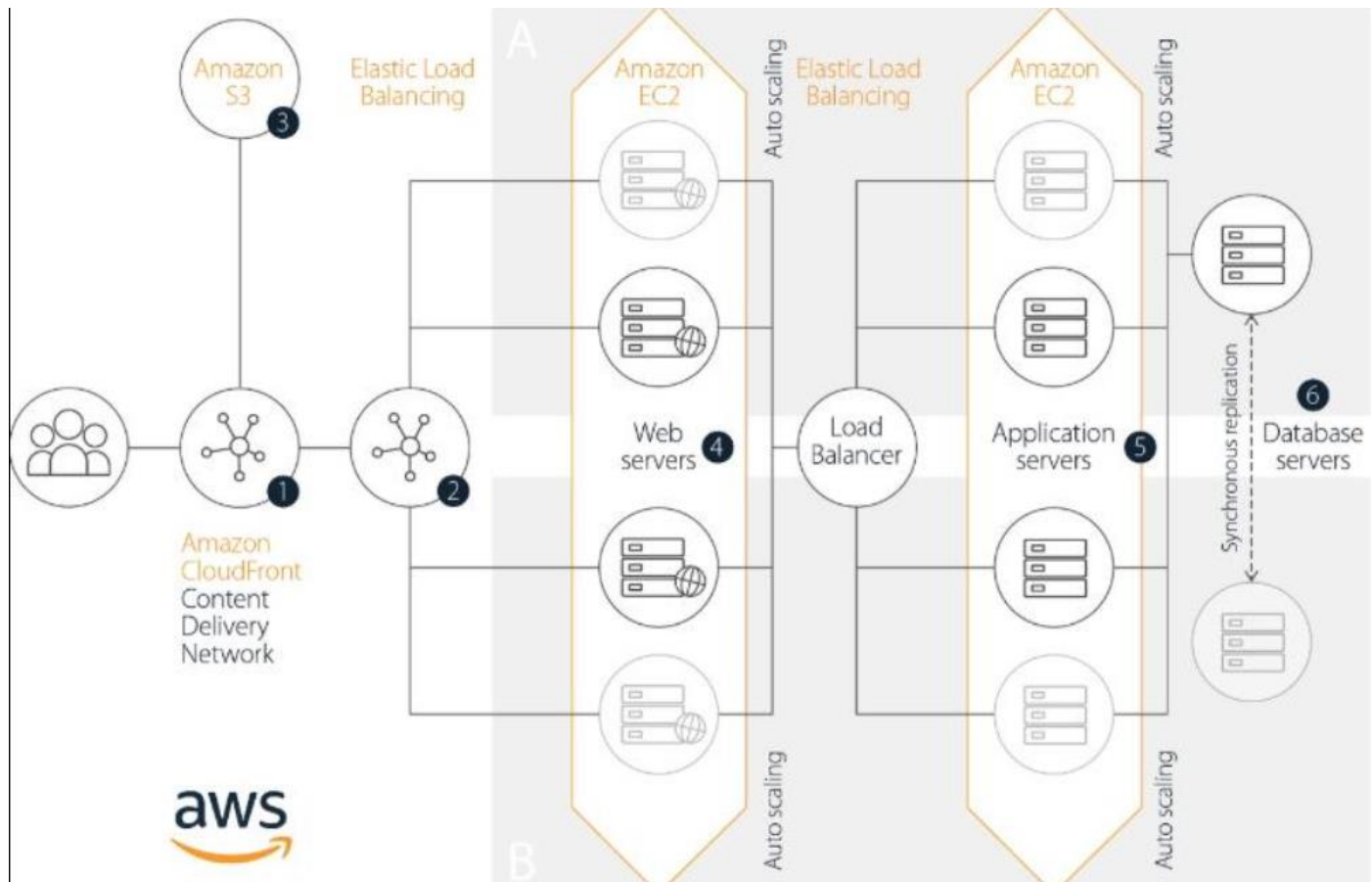
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Q1. Cloud Architecture for Video Streaming Platform

Architecture :



- Use CDN (Content Delivery Network) to cache videos near users.
- Store videos in cloud object storage (Amazon S3 / Azure Blob / Google Cloud Storage).
- Place load balancers in front of streaming servers.
- Use auto-scaling groups to add or remove servers based on traffic.
- Use distributed databases and monitoring services.

Load Balancing & Auto-Scaling

- Load balancer distributes requests evenly across servers.

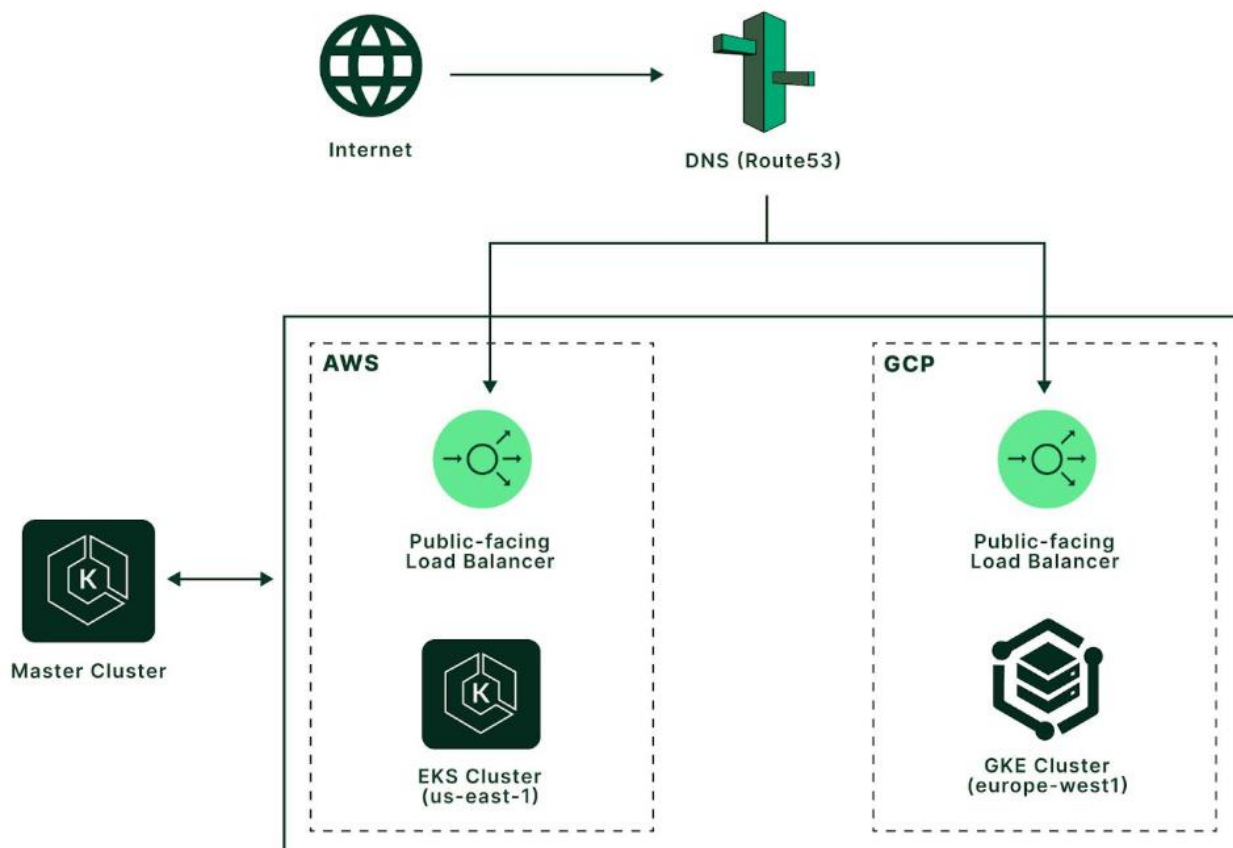
- Auto-scaling increases instances during peak demand and decreases them during low traffic.
- This reduces buffering and improves user experience.

Cost Optimization

- Use CDN caching to reduce bandwidth costs.
- Store old videos in cold/archive storage.
- Use spot/reserved instances for predictable workloads.
- Enable automatic scaling down when traffic decreases.

Q2. Multi-Cloud Container Deployment

Proposed Solution :



- Use Kubernetes for container orchestration.
- Deploy clusters across AWS, Azure, and GCP.
- Use Helm or GitOps (ArgoCD/Flux) for consistent deployments.

Service Discovery & Scaling

- Kubernetes DNS and Services provide service discovery.
- Use Horizontal Pod Autoscaler (HPA) for automatic scaling.
- Use Ingress controllers for external access.

Benefits

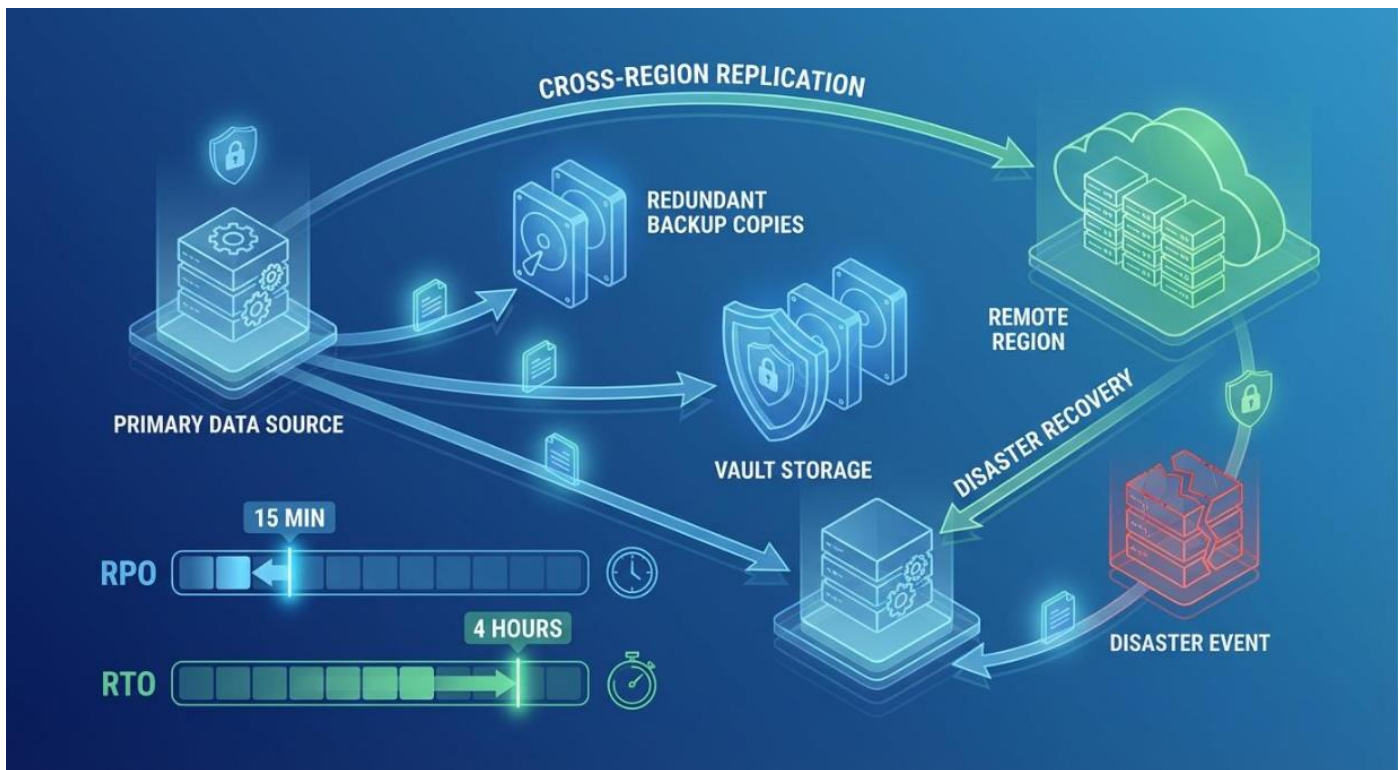
- Avoids vendor lock-in.
- Higher availability.
- Better geographic coverage.

Risks

- Increased management complexity.
- Higher networking cost.
- Security and compliance challenges.

Q3. Backup and Disaster Recovery for E-Learning Platform

Solution :



- Enable object versioning for files.
- Take automated database snapshots.
- Use cross-region replication.
- Store backups in a separate backup account/project.

RTO and RPO

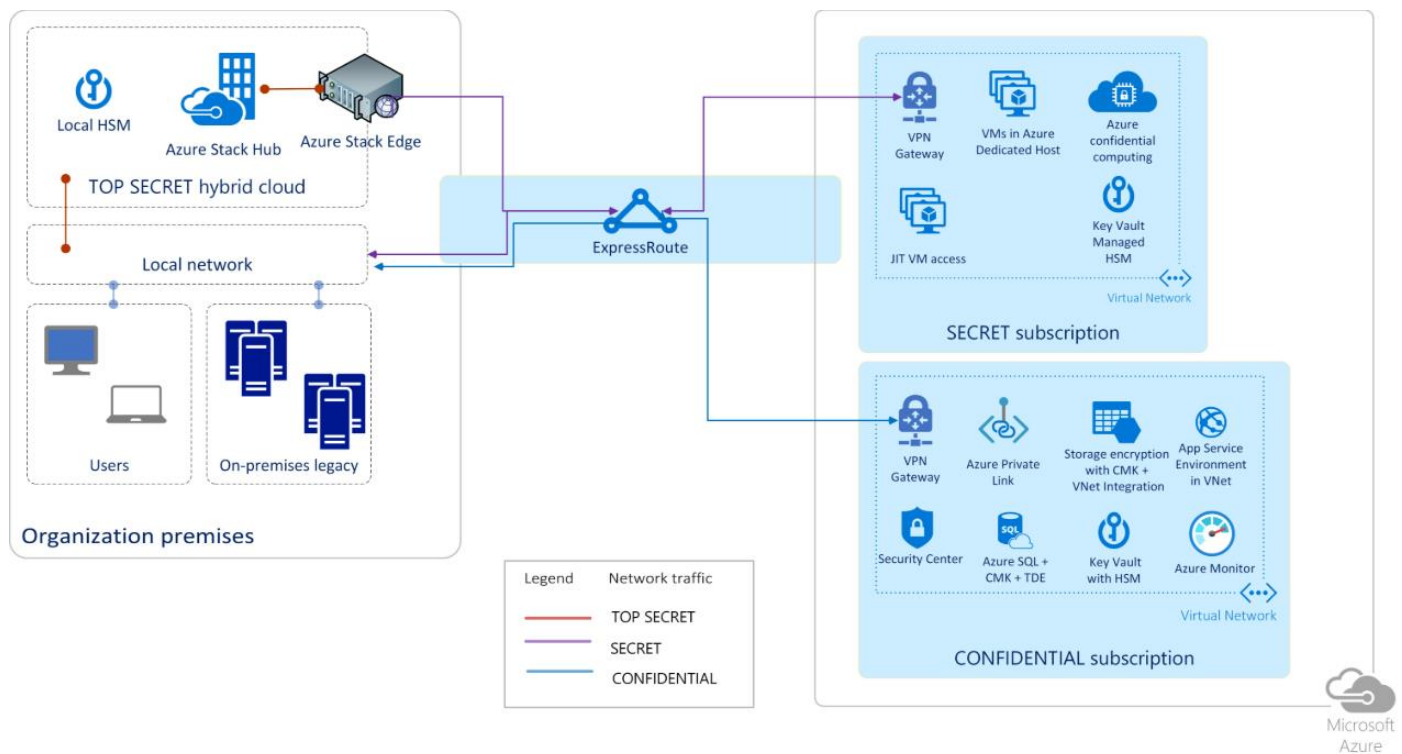
- RTO: Use warm standby or replicated systems for faster recovery.
- RPO: Frequent backups and continuous replication minimize data loss.

Business Continuity Steps

- Detect failure.
- Switch to replica/DR site.
- Restore latest backup if required.
- Validate services.
- Resume normal operations.

Q4. Hybrid Cloud for Government Agency

Hybrid Architecture :



- Keep sensitive applications on-premises.
- Host non-sensitive workloads in the public cloud.
- Connect environments using VPN or dedicated private link

Identity Management

- Use centralized IAM/Active Directory.
- Enable single sign-on (SSO) and multi-factor authentication (MFA).

Encryption

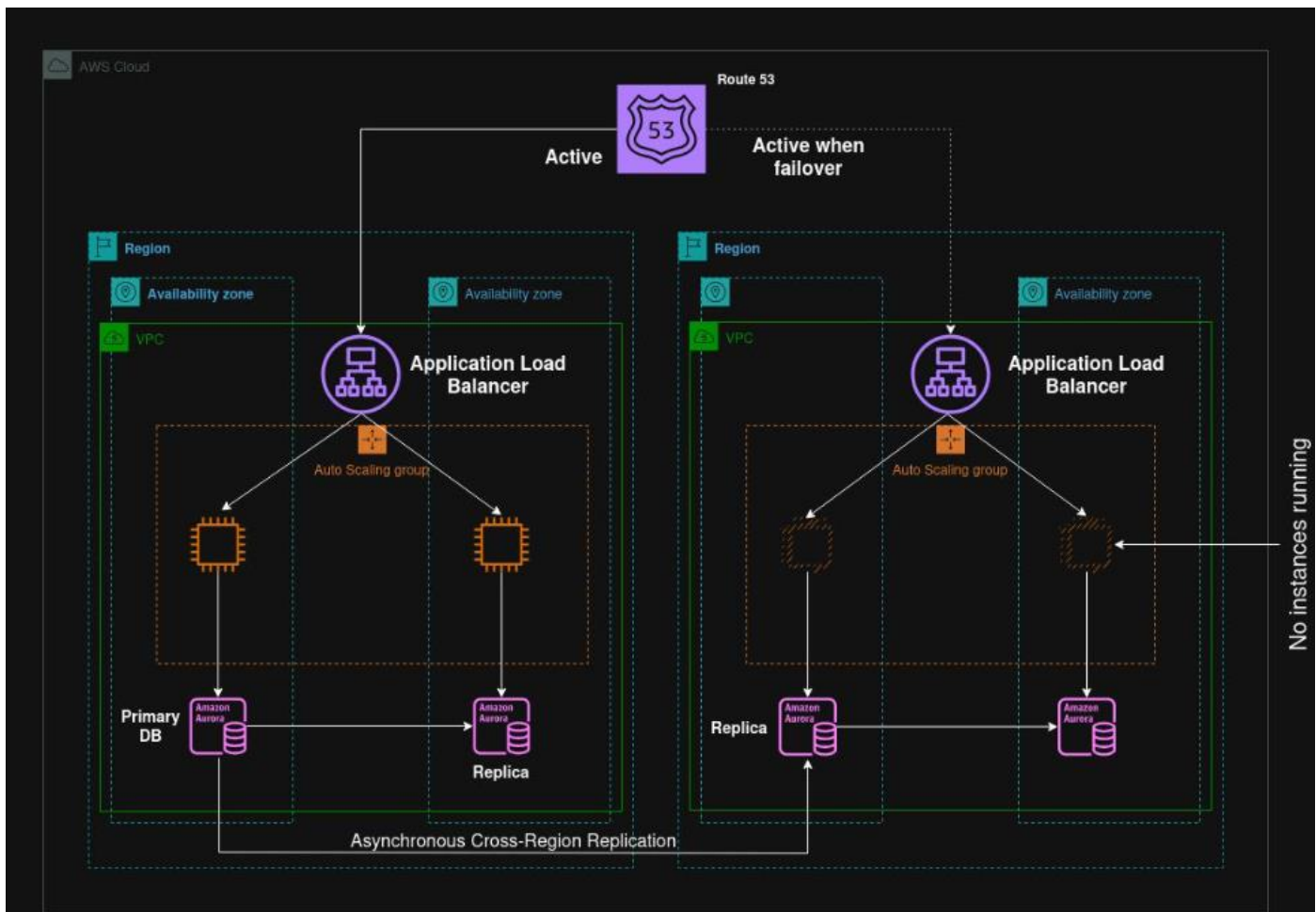
- Encrypt data in transit (TLS) and at rest (AES-256).
- Use cloud key management services.

Challenges

- Complex management.
- Latency between environments.
- Compliance monitoring.
- Consistent security policies.

Q5. Multi-Region High Availability Strategy

Deployment Strategy :



- Deploy the application in two or more regions.
- Use global load balancer / DNS routing.
- Prefer active-active for critical SaaS applications.

Data Replication & Failover

- Use cross-region database replication.
- Configure automatic failover.
- Regularly test disaster recovery procedures.

Monitoring & Alerting

- Monitor CPU, memory, latency, errors, and replication lag.
- Use cloud monitoring tools (CloudWatch, Azure Monitor, GCP Monitoring).
- Configure alerts through email, SMS, or incident tools.